CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

10.2 Kirchhoff's Laws

Syllabus Section: 10. D.C. Circuits **Paper Type:** Structured Questions (Paper 2) **Total:** 7 Questions (91 Marks)

Question 1 Easy [11 marks]

Part (a) [3 marks]

Three resistors of resistances R_1, R_2 and R_3 are connected as shown in Fig. 1.1.

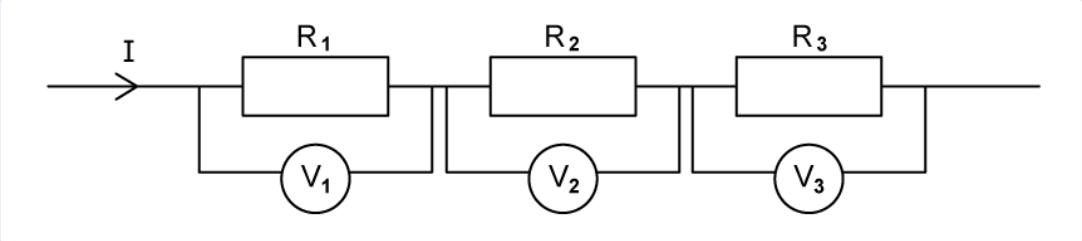


Fig 1.1

The potential difference across the resistors are V_1, V_2 and V_3. The current in the combination of resistors is *I*.

Show that the total resistance *R* of the combination is given by the equation.

$$R = R_1 + R_2 + R_3$$

Mark Scheme (3 marks):

The total resistance *R* is given by the equation $R = R_1 + R_2 + R_3$ because:

- The total potential difference in the circuit, *V* is equal to the sum of the potential differences across each resistor:
  - $V = V_1 + V_2 + V_3$ *[1 mark]*

Consider Ohm's Law and substitute into the expression for current, *V*:

- $IR = IR_1 + IR_2 + IR_3$ *[1 mark*
- $R = R_1 + R_2 + R_3$ *[1 mark]*

[Total: 3 marks]

Part (b) [3 marks]

Three resistors are placed in a circuit with a cell of 10 V and negligible internal resistance and two switches S_1 and S_2. This is shown in Fig 1.1.

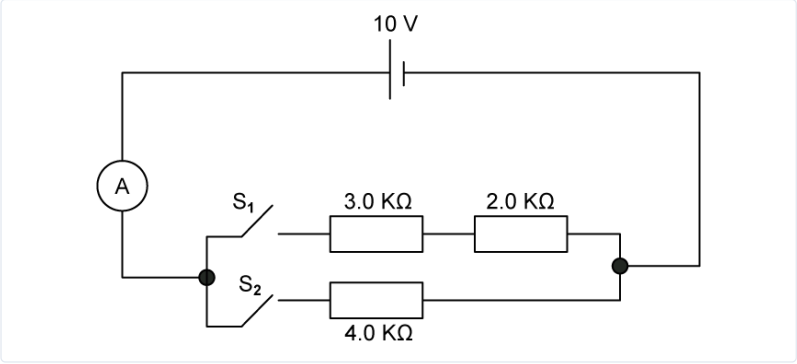


Fig 1.1

Complete table 1.1.

Table 1.1

Position of switches		
S_1	S_2	Total resistance / kΩ
open	closed	
closed	open	

closed	closed	

Mark Scheme (3 marks):

A complete Table 1.1 should look like:

Position of switches		
S_1	S_2	Total resistance / kΩ
open	closed	***4.0***
closed	open	***5.0***
closed	closed	***2.2***

One mark for each correct row

[Total: 3 marks]

If you're struggling with the total resistances when both switches are closed, split the circuit into parallel and series sections.

1. *First, combine the series resistors. This is 3.0 + 2.0 = 5.0 kΩ. This acts like 1 resistor with a resistance of 5.0 kΩ.*
2. *This is now in parallel with the 4.0 kΩ resistor which is calculated in parallel. i.e.* $\frac{1}{R_T} = \frac{1}{5.0} + \frac{1}{4.0} = 0.45$. $R_T = \frac{1}{0.45} = 2.2$ kΩ

Part (c) **[2 marks]**

The cell has an e .m .f of 10 V and negligible internal resistance.

Calculate the current from the cell when

(i) switch S_1 is closed,

[1]

(ii) both S_1 and S_2 are closed.

[1]

Mark Scheme (2 marks):

The current when

(i) switch S_1 is closed is:

List the known quantities:

- Total resistance, R = 5.0 kΩ = 5.0 × 10^3 Ω (from part b)

Calculate the current:

- $V = IR \Rightarrow I = \frac{V}{R}$
- $I = \frac{10}{5.0 \times 10^3} = 2 \times 10^{-3}$ A 2 mA *[1 mark]*

(ii) switch S_1 and S_2 are closed is:

List the known quantities:

- Total resistance, R = 2.2 kΩ = 2.2 × 10^3 Ω (from part b)

Calculate the current:

- $I = \frac{10}{2.2 \times 10^3} = 4.545 \times 10^{-3} = 4.5 \times 10^{-3}$ A 4.5 mA *[1 mark]*

[Total: 2 marks]

Due to Kirchhoff's second law, the total potential difference across the branch with S_1 and S_2 is the same. Therefore, V in the equation for both is 10 V.

Part (d) **[3 marks]**

Using your answers in part (c), determine the current from the cell if just switch S_2 is closed.

Mark Scheme (3 marks):

The current from the cell if switch S_2 is closed is:

List the known quantities:

- The current when S_1 is closed, $I_1 = 2.0 \times 10^{-3}$ A (from part c)
- The current when S_1 and S_2 is closed, $I = 4.5 \times 10^{-3}$ A (from part c)

Calculate the current when S_2 is closed, I_2:

- Kirchhoff's first law: $I = I_1 + I_2$ *[1 mark]*
- $I_2 = I - I_1 = 4.5 \times 10^{-3} - 2.0 \times 10^{-3}$ *[1 mark]*
- $I_2 = 2.5 \times 10^{-3}$ A 2.5 mA *[1 mark]*

[Total: 3 marks]

Although you can get this same answer by doing $V = \frac{I}{R} = \frac{10}{4.0 \times 10^{3}} = 2.5 \times 10^{-3}$ V*, the question specifically states by using your answers to part (c) – this implies Kirchhoff's first law. You can, of course, check your answer with Ohm's law, but you cannot gain any marks for reaching an answer this way.*

Question 2 Easy [12 marks]

Part (a) [2 marks]

(i) State Kirchhoff's first law.

[1]

(ii) State the quantity that is conserved by this law.

[1]

Mark Scheme (2 marks):

(i) Kirchhoff's first law states that:

- The sum of currents into a junctions equals the sum of the currents out of a junction; *[1 mark]*

(ii) The quantity conserved by this law is:

- Charge; *[1 mark]*

[Total: 2 marks]

*It is common to mistakenly say that 'current' is conserved in this law. Although that's true, current is just the flow of **charge**, which is the fundamental conservation linked with Kirchhoff's first law. Therefore, for part (ii), 'current' will **not** get the mark.*

Part (b) [2 marks]

(i) State Kirchhoff's second law.

[1]

(ii) State the quantity that is conserved by this law.

[1]

Mark Scheme (2 marks):

(i) Kirchhoff's second law states that:

- The sum of e.m.f's equals the sum of the p.d's around a loop / circuit; *[1 mark]*

(ii) The quantity conserved by this law is:

- Energy; *[1 mark]*

[Total: 2 marks]

*It is common to mistakenly say that 'p.d' is conserved in this law. However, p.d. is the measure of energy per unit charge transferred between two points in a circuit. Therefore, it is technically **energy** that is conserved.*

Part (c) [6 marks]

Fig 1.1 shows a list of circuit components.

Earth	
Variable resistor	
Heater	
LED	
Thermistor	
Microphone	

Fig 1.1

In the right hand column, sketch the circuit symbol for the component in the left hand column of Fig 1.1.

Mark Scheme (6 marks):

The correct circuit symbols are:

Earth	
Variable resistor	
Heater	
LED	
Thermistor	
Microphone	

- *One mark for each correct circuit symbol*

[Total: 6 marks]

*There are some niche circuit symbols required in this question, however, they are all required according to the CIE IAL syllabus. The end of the syllabus has a list of all the circuit symbols that you should recognise in your exam questions, make sure to familiarise yourself with **all** of them.*

Part (d) [2 marks]

Two cells of e.m.f. E_1 and E_2 are negligible internal resistance are connected to resistors R in a circuit shown in Fig 1.2.

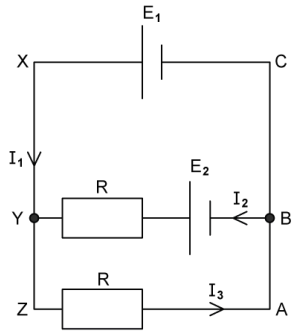


Fig 1.2

Use Kirchhoff's laws to state the relation between

(i) I_1, I_2 and I_3

[1]

(ii) E_2, R, I_2 and I_3 in loop YZABY.

[1]

Mark Scheme (2 marks):

The relation between:

(i) I_1, I_2 and I_3 is:

- $I_1 + I_2 = I_3$ *[1 mark]*

(ii) E_2, R, I_2 and I_3 in loop YZABY is:

- $E_2 = I_2R + I_3R$ *[1 mark]*

[Total: 2 marks]

Remember from Ohm's law that V = IR.

Part (i) is using Kirchhoff's first law, whilst part (ii) is using Kirchhoff's second law.

Question 3 Medium [12 marks]

Part (a) [2 marks]

Three resistors of resistances R_1, R_2 and R_3 are connected as shown in Fig. 1.1.

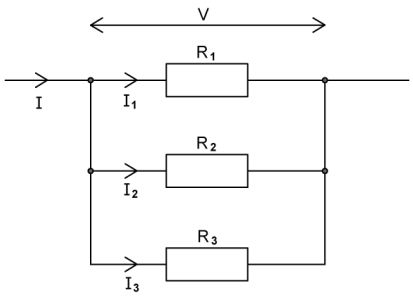


Fig. 1.1

The currents in the resistors are I_1, I_2 and I_3. The total current in the combination of resistors is I and the potential difference across the combination is V.

Show that the total resistance R of the combination is given by the equation

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Mark Scheme (2 marks):

Derive the expression for total resistance R in parallel:

- The total current in the circuit I is equal to the sum of the currents through each resistor:
  - $I = I_1 + I_2 + I_3$ *[1 mark]*

Consider Ohm's Law and substitute into the expression for current, I:

- $\frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$

  AND

- $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$ *[1 mark]*

[Total: 2 marks]

- *The potential difference across the parallel combination is V, so:*
  - $V_1 = V_2 = V_3 = V$

Part (b) [10 marks]

A battery of electromotive force (e.m.f.) 6.0 V and internal resistance r is connected to an external resistor of resistance 12 Ω and a thermistor X, as shown in Fig. 1.2.

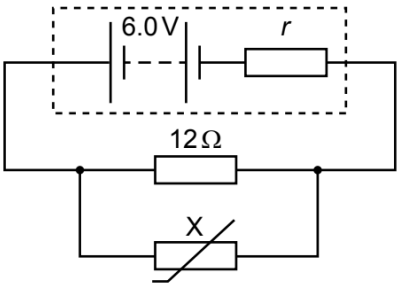


Fig. 1.2

(i) By considering energy, explain why the potential difference across the terminals of the battery is less than the e.m.f.

[1]

(ii) A charge of 2.5 kC passes through the battery.

Calculate:

- the total energy transferred by the battery

energy = ... J

- the number of electrons that pass through the battery.

number = ...

[3]

(iii) The combined resistance of the external resistor and thermistor X connected in parallel is 4.8 Ω.

Calculate the resistance of X.

resistance = ... Ω [1]

(iv) Use your answer in **(b)(iii)** to determine the ratio

$$\frac{\text{power dissipated in thermistor X}}{\text{power dissipated in 12 Ω resistor}}$$

ratio = ... [2]

(v) The temperature of thermistor X is now decreased.

State and explain the effect, if any, of this temperature change on the total power produced by the battery.

[3]

Mark Scheme (10 marks):

List the known quantities:

- e.m.f., V = 6.0 V
- Charge, Q = 2.5 kC = 2.5 × 10^3 C
- Elementary charge, e = 1.6 × 10^{-19} C
- Combined resistance, R = 4.8 Ω
- Resistance of resistor, R_1 = 12 Ω

(i) Explain why the potential difference across the terminals of the battery is less than the e.m.f:

- The energy (inside the battery) is dissipated in the internal resistance; *[1 mark]*

(ii) Calculate:

- the total energy transferred by the battery:
  - $V = \frac{W}{Q}$
  - $W = VQ$ *[1 mark]*
  - $W = 6 \times 2.5 \times 10^3$
  - $W = 1.5 \times 10^4$ J *[1 mark]*

- the number of electrons that pass through the battery:
  - $number\ of\ electrons = \frac{Q}{e}$
  - $number\ of\ electrons = \frac{2.5 \times 10^3}{1.6 \times 10^{-19}}$
  - $number\ of\ electrons = 1.56 \times 10^{22} = 1.6 \times 10^{22}$ $2\ s.f.$ *[1 mark]*

(iii) Calculate the resistance of X:

- $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$
- $\frac{1}{4.8} = \frac{1}{12} + \frac{1}{R_X}$
- $\frac{1}{R_X} = \frac{1}{8}$
- $R_X = 8\ \Omega$ *[1 mark]*

(iv) Determine the ratio $\frac{\text{power dissipated in thermistor X}}{\text{power dissipated in 12 Ω resistor}}$:

- $P = \frac{V^2}{R}$ *[1 mark]*
- $P_X = \frac{6^2}{8} = 4.5$ W

- $P_1 = \frac{6^2}{12} = 3$ W
- $\frac{P_X}{P_1} = \frac{4.5}{3} = 1.5$ *[1 mark]*

(v) As the temperature of the thermistor decreases:

- The resistance increases; *[1 mark]*
- This means the total resistance (in the circuit) increases ***OR*** the total current (in the battery) decreases; *[1 mark]*
- (Hence) the power decreases; *[1 mark]*

[Total: 10 marks]

- *In part (v)* $P = \frac{V^2}{R}$ *so when* R *increases* P *decreases*

Question 4 Medium [15 marks]

Part (a) [1 mark]

A student investigates the potential difference in a circuit. The circuit is set up as shown in Fig. 2.1.

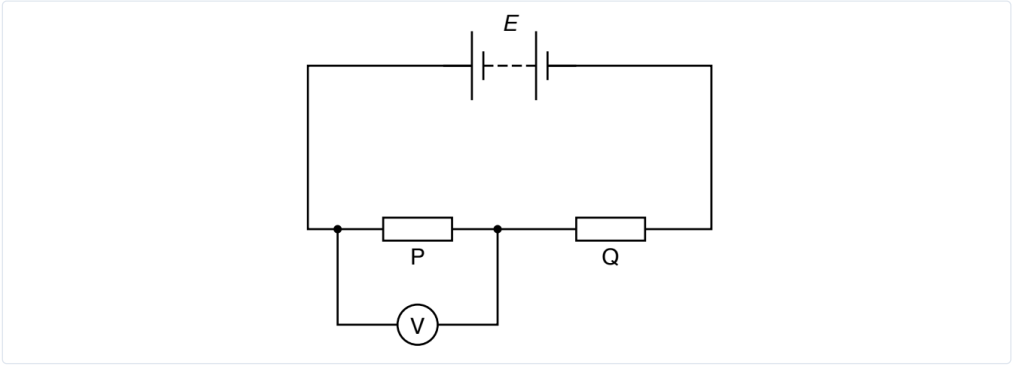


Fig. 2.1

Two resistors P and Q are connected in series to a power supply of electromotive force (e.m.f.) E and negligible internal resistance. Resistor P has resistance P.

The potential difference V across resistor P is measured. The experiment is repeated for different values of P.

It is suggested that V and P are related by the equation

$$V = \frac{P}{P+Q}E$$

where Q is the resistance of resistor Q. The value of Q is kept constant.

A graph is plotted of $\frac{1}{V}$ on the y-axis against $\frac{1}{P}$ on the x-axis.

Determine expressions for the gradient and the y-intercept.

gradient = ...

y-intercept = ...

Mark Scheme (1 marks):

Determine expressions for the gradient and the y-intercept:

- $V = \frac{P}{P+Q}E = \frac{PE}{P+Q} \Rightarrow \frac{1}{V} = \frac{P+Q}{PE} = \frac{1}{E} + \frac{Q}{PE}$
- $\frac{1}{V} = \frac{1}{E} + \frac{Q}{PE} \Rightarrow \frac{1}{V} = (\frac{Q}{E})\frac{1}{P} + \frac{1}{E}$ y =

Determine the gradient and y-intercept:

- Hence, the gradient = $\frac{Q}{E}$ **AND** y-intercept = $\frac{1}{E}$ *[1 mark]*

[Total: 1 mark]

Part (b) [2 marks]

Values of P, V and $\frac{1}{V}$ are given in Table 2.1.

Table 2.1

P / Ω	V / V	$\frac{1}{P}$ / 10^{-3} Ω^{-1}	$\frac{1}{V}$ / V^{-1}
250	0.66		1.52
330	0.86		1.16
470	1.15		0.870
560	1.30		0.769
680	1.49		0.671

840	1.64		0.610

Each value of *P* has an uncertainty of ±10%.

Calculate and record values of $\frac{1}{P}$ / 10^{-3} Ω^{-1} in Table 2.1.

Include the absolute uncertainties in $\frac{1}{P}$.

Mark Scheme (2 marks):

Calculate and record values of $\frac{1}{P}$ / 10^{-3} Ω^{-1} in Table 2.1.

Table 2.1

P / Ω	V / V	$\frac{1}{P}$ / 10^{-3} Ω^{-1}	$\frac{1}{V}$ / V^{-1}
250	0.66	4.0 ± 0.40	1.52
330	0.86	3.0 ± 0.30	1.16
470	1.15	2.1 ± 0.21	0.870
560	1.30	1.8 ± 0.18	0.769
680	1.49	1.5 ± 0.15	0.671
840	1.64	1.2 ± 0.12	0.610

- All values and uncertainties correct; *[1 mark]*
- All values and uncertainties quoted to 2 significant figures; *[1 mark]*

[Total: 2 marks]

Don't be fooled by the equation. Calculating $\frac{1}{P}$ is simple, just find the reciprocal of values of P from ***column 1****.*

P is given to ***2 s.f****. so $\frac{1}{P}$ should also be given as the same*

The ***absolute uncertainty*** *of $\frac{1}{P}$ is calculated by finding* ***10%*** *of each value of $\frac{1}{P}$*

Part (c) **[8 marks]**

(i) Plot a graph of $\frac{1}{V}$ / V^{-1} against $\frac{1}{P}$ / 10^{-3} Ω^{-1}.

Include error bars for $\frac{1}{P}$.

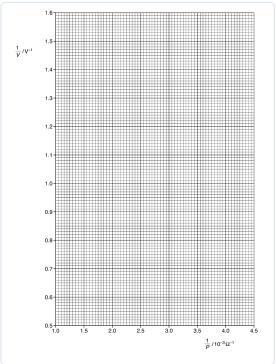
[2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines.

[2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = ... [2]



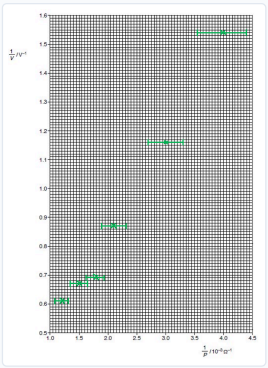
(iv) Determine the y-intercept of the line of best fit. Include the absolute uncertainty in your answer.

y-intercept = ... [2]

Mark Scheme (8 marks):

(i) Plot a graph of 1/V against 1/P:

A graph that would score 2 marks should look like this:



Plotted points:

*All **three** for 1 mark:*

- All six points from **(b)** plotted correctly
- Each point must be accurate to within half a small square
- The diameter of the point must be less than half a small square; *[1 mark]*

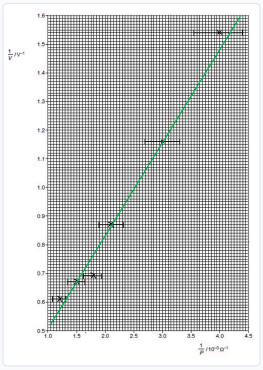
Error bars:

*All **three** for 1 mark:*

- All uncertainties for $\frac{1}{P}$ are plotted correctly as error bars
- All error bars are plotted
- The length of the bar must be accurate to less than half a small square and symmetrical; *[1 mark]*

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines:

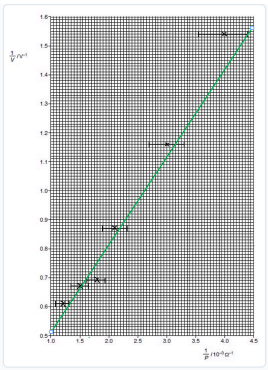
A graph that would score 1 mark should look like this:



Straight line of best fit drawn:

- Lower end of line must pass between (1.50, 0.70) and (1.65, 0.70) and upper end of line must pass between (3.60, 1.40) and (3.80, 1.40); *[1 mark]*

A graph that would score 1 mark should look like this:

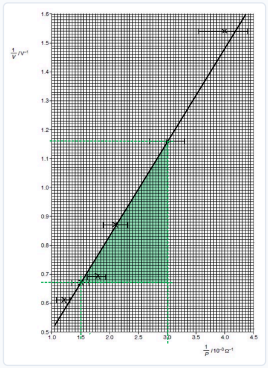


Worst acceptable line drawn:

- Steepest possible line that passes through all the error bars ***AND*** all error bars must be plotted; *[1 mark]*

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

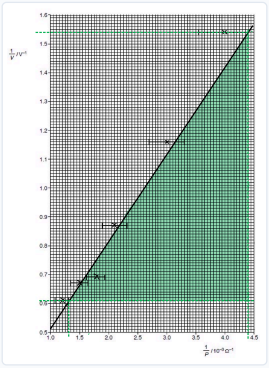
Draw lines on the graph to obtain a gradient triangle for the line of best fit:



Calculate the gradient of the best fit line:

- Point 1: (1.5, 0.671)
- Point 2: (3.0, 1.16)
- $gradient = m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1.16 - 0.671}{3 - 1.5}$ = 0.326 *[1 mark]*

Draw lines on the graph to obtain a gradient triangle for the line of worst fit:



Calculate the gradient of the worst fit line:

- Point 1: (1.3, 0.61)
- Point 2: (4.4, 1.54)
- $gradient = m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1.54 - 0.61}{4.4 - 1.3}$ = 0.3

Calculate the uncertainty in the gradient of the best fit line:

- Uncertainty = gradient of best fit line − gradient of worst fit line
- Uncertainty = 0.326 − 0.3

- Uncertainty = ±0.026 *[1 mark]*

(iv) Determine the y-intercept of the line of best fit. Include the absolute uncertainty in your answer:

List the known quantities:

Line of best fit:

- Point 1: (1.5, 0.671)
- Gradient, m = 0.326

Line of worst fit:

- Point 1: (1.3, 0.61)
- Gradient, m = 0.3

Substitute quantities from point 1 on the line of best fit into the standard equation of a straight line:

- $y = mx + c$
- $y - mx = c$
- $0.671 - 0.326 \times 1.5 = c$
- y-intercept for the line of best fit, $c = 0.182 \ \text{V}^{-1}$ *[1 mark]*

Substitute quantities from point 1 on the line of worst fit into the standard equation of a straight line:

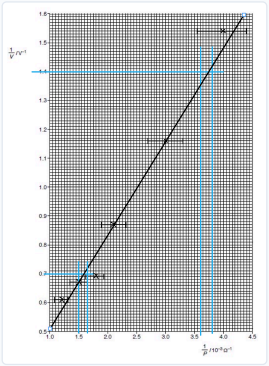
- $y - mx + c$
- $0.61 - 0.3 \times 1.3 = c$
- y-intercept for line of worst fit, $c = 0.22 \ \text{V}^{-1}$

Calculate the uncertainty in the y-intercept:

- Uncertainty = y-intercept of best-fit line − y-intercept of worst-fit line
- Uncertainty = 0.182 − 0.22
- Uncertainty = ±0.038 *[1 mark]*

[Total: 8 marks]

- *It is **difficult** to draw the line of best fit with an **equal** number of **points above and below**.*
- *An **attempt** to do this and have the line between the coordinates indicated will award you the **mark***



- *When drawing a gradient triangle it is best to use **plotted points** that lie **exactly** on the line of best fit*
- *There is a range within which a **line of best fit is accepted**, so your **gradient and y-intercept values** will be **different** to those calculated.*
- *Not many students will draw exactly the same line of best fit.*
- *If you have followed the **same steps as above,** even though your **values are different**, you will be awarded the marks*

Part (d) **[4 marks]**

(i) Using your answers to **(a), (c)(iii)** and **(c)(iv)**, determine the values of E and Q. Include appropriate units.

E = ...

Q = ...

[2]

(ii) Determine the percentage uncertainty in *E*.

percentage uncertainty in *E* = ... % [1]

(iii) Determine the absolute uncertainty in *Q*.

absolute uncertainty in *Q* = ... [1]

Mark Scheme (4 marks):

(i) Using your answers to **(a), (c)(iii)** and **(c)(iv)**, determine the values of E and Q. Include appropriate units:

List the known quantities:

- y-intercept = $\frac{1}{E}$ (from part (a))
- y-intercept, c = 0.182 V^{-1} (from part b, iv)
- Gradient = $\frac{Q}{E}$ (from part (a))
- Gradient, m = 0.326 (from part b, iii)

Use the expression for y-intercept and the known quantity to calculate E:

- $E = \frac{1}{c}$
- $E = \frac{1}{0.182}$
- $E = 5.49$ V *[1 mark]*

Use the expression for gradient and the known quantities to calculate, Q:

- $Q = m \times E$
- $Q = 0.326 \times 5.49$
- $Q = 1.79$ Ω *[1 mark]*

(ii) Determine the percentage uncertainty in *E*:

List the known quantities:

- y-intercept, c = 0.182
- Uncertainty in y-intercept, Δc = 0.038

Calculate the percentage uncertainty in y:

- % uncertainty in *E* = % uncertainty in y-intercept
- % uncertainty in y-intercept = $\frac{\Delta c}{c} \times 100$
- % uncertainty = $\frac{0.038}{0.182} \times 100$
- % uncertainty = 20.8 % = 21 % (2s.f.) *[1 mark]*

(iii) Determine the absolute uncertainty in *Q*:

List the known quantities:

- Absolute uncertainty in gradient, Δm = ±0.026
- Gradient of best-fit line, m = 0.326
- % uncertainty in *E* = % uncertainty in y-intercept = 21 %

Calculate the % uncertainty in the gradient of the best flt line:

- % uncertainty = $\frac{\Delta m}{m} \times 100$
- % uncertainty = $\frac{0.026}{0.326} \times 100$
- % uncertainty = 7.98 % = 8.0 % (2s.f.)

Calculate the % uncertainty in *Q*:

- % uncertainty in *Q* = % uncertainty in *E* + % uncertainty in gradient
- % uncertainty in *Q* = 21 + 8 = 29%

Calculate the absolute uncertainty in *Q*:

- Absolute uncertainty = $\frac{percentage\ \ uncertainty\ \ in\ \ Q}{100} \times Q$
- Absolute uncertainty = $\frac{29}{100} \times 1.79$

- Absolute uncertainty = ±0.5191 Ω = ±0.52 Ω (2s.f.) *[1 mark]*

[Total: 4 marks]

- *You may obtain **slightly different values** than those above as a result of:*
  - *A different **line of best fit***
  - *A different **line of worst fit***
  - *Giving a **different gradient***
  - *And a **different y-intercept***
- *If you have followed the **same steps** you will be awarded the marks*

- *Remember when **multiplying / dividing quantities** you must **add the uncertainties** to combine them*
- *Converting from % to absolute uncertainty requires the inverse % to be found:*

$$\%\ uncertainty = \frac{absolute}{quantity} \times 100$$

$$\frac{\%\ uncertainty \times quantity}{100} = absolute$$

Question 5 Medium [12 marks]

Part (a) [4 marks]

The circuit shown in Fig. 1.1 contains a battery with an e.m.f. E and negligible internal resistance r connected to four resistors R_1, R_2, R_3 and R_4, each of resistance R.

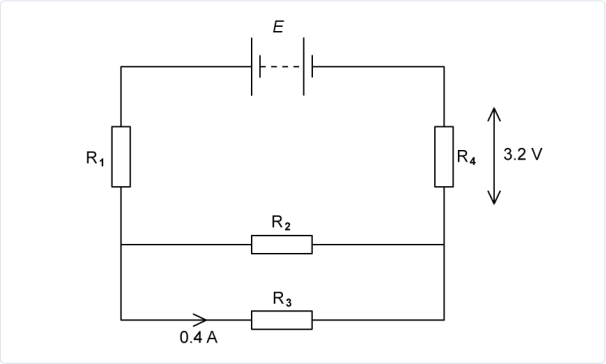


Fig. 1.1

The current in R_3 is 0.40 A and the potential difference across R_4 is 3.2 V.

Determine the resistance R of each of the fixed resistors.

Mark Scheme (4 marks):

List the known quantities:

- Current in R_2 and R_3 = 0.40 A
- P.d. across R_1 and R_4 = 3.2 V

Determine the unlabelled values of current and potential difference:

- Current in R_1 and R_4 = 2 × 0.40 = 0.80 A *[1 mark]*
- P.d. across R_2 and R_3 = $\frac{3.2}{2}$ = 1.6 V *[1 mark]*

Calculate the resistance R_1 and R_4:

- Recall: $V = IR \rightarrow R = \frac{V}{I}$
- $R = \frac{3.2}{0.80} = 4\ \Omega$ *[1 mark]*

Calculate the resistance R_2 and R_3:

- Hence, $R = \frac{1.6}{0.40} = 4\ \Omega$ *[1 mark]*

[Total: 4 marks]

The question tells you that R should be the same for all the resistors, so calculate R for all of them just to check.

Part (b) [2 marks]

Determine the e.m.f. of the battery.

Mark Scheme (2 marks):

List the known quantities:

- P.d. across R_1 and R_4 = 3.2 V (From part (a))
- P.d. across R_2 and R_3 = 1.6 V (From part (a))

Calculate the total p.d. in the circuit:

- Total p.d. = e.m.f.
- e.m.f. = p.d. across R_1 + p.d. across R_2 + p.d. across R_3 + p.d. across R_4
- e.m.f. = 3.2 + 1.6 + 3.2 + 1.6 *[1 mark]*
- e.m.f. = 9.6 V *[1 mark]*

[Total: 2 marks]

This question can also be solved by finding the total resistance of the combination and using the equation:

$$V = IR$$

Part (c) **[3 marks]**

A fifth resistor R_5 also with resistance R is now added in parallel to R_2 and R_3. The current through R_3 remains at 0.40 A.

Determine the total current in this circuit. Explain your answer.

Mark Scheme (3 marks):

List the known quantities:

- Current through R_3 = 0.40 A

Total current through R_1 and R_4:

- Total current = 0.40 + 0.40 + 0.40 = 1.2 A *[1 mark]*

This is because:

- According to Kirchhoff's first law: The sum of the currents entering a junction is always equal to the sum of the currents out of the junction; *[1 mark]*
- In this circuit the current through each branch of the parallel components of the circuit is the same because each resistor has a resistance R; *[1 mark]*

[Total: 3 marks]

Part (d) **[3 marks]**

The battery in the circuit is now replaced by one with the same e.m.f. but internal resistance is not negligible.

Explain whether this will change the total power produced by the battery.

Mark Scheme (3 marks):

This will change the total power produced by the battery because:

- The total resistance in the circuit has increased; *[1 mark]*
- So the current in the battery decreases; *[1 mark]*
- Hence, the total power decreases; *[1 mark]*

[Total: 3 marks]

Consider the equations:

$$V = IR$$

When R increases then I decreases for the battery to provide a constant p.d.

$$P = IV$$

So an increase in I results in a decrease in P.

Question 6 Hard [14 marks]

Part (a) [4 marks]

A circuit containing a junction P has two power supplies with e.m.f 6.0 V and 2.0 V and two resistors with resistance 30 Ω and 10 Ω is shown in Fig. 1.1.

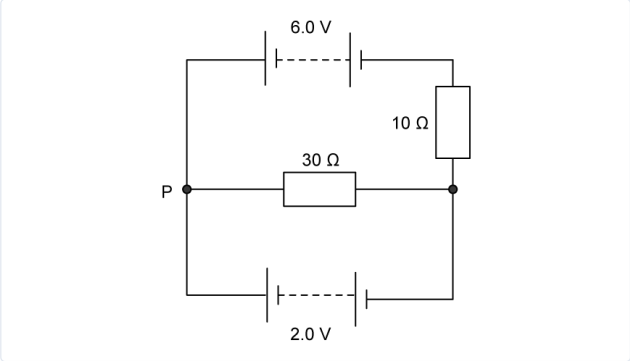


Fig. 1.1

Draw on the diagram to label the currents in each branch of the circuit and obtain an expression for the current at P.

Mark Scheme (4 marks):

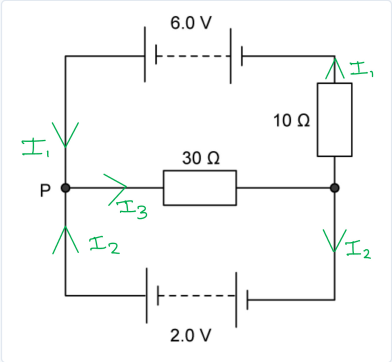
List the known quantities:

- e.m.f-1 = 6.0 V
- e.m.f-2 = 2.0 V
- Resistance-1 = 10 Ω
- Resistance-2 = 30 Ω

Consider the circuit diagram and identify currents I_1, I_2 and I_3:

A diagram that would score 3 marks should look like this:

- To determine the direction of the current first identify the positive and negative terminals of each power supply



- Current I_1 is identified as both going into P from the positive side of the 6.0 V power supply and in the same branch as the 10 Ω resistor; *[1 mark]*
- Current I_2 is identified as both going into P from the positive side of the 2.0 V power supply and going into the negative side of the 2.0 V power supply; *[1 mark]*
- Current I_3 is identified as going out of P in the 30 Ω resistor branch; *[1 mark]*

Apply Kirchoff's first law for the current to obtain an expression for the current at P:

- Current at P = $I_3 = I_1 + I_2$ (Equation 1)

  OR

- If currents have been labelled the opposite way around to the diagram above: Current at P = $I_1 = I_2 + I_3$ *[1 mark]*

[Total: 4 marks]

Remember that current comes from the positive side of the battery. It can be helpful to mark the currents for each power supply and each branch. The sign convention shown in the solution makes the most sense but it is also possible to note the currents in the opposite directions and obtain the second equation. As long as you are consistent it does not matter.

Part (b) [4 marks]

Determine the values of the currents at P for the circuit in part a).

Mark Scheme (4 marks):

List the known quantities:

- e.m.f-1 = 6.0 V
- e.m.f-2 = 2.0 V
- Resistance-1 = 10 Ω
- Resistance-2 = 30 Ω
- Current at P = $I_3 = I_1 + I_2$ (Equation 1) (from part a)

Apply Kirchoff's second law for the potential difference in the top loop of the circuit:

- Recall Ohm's Law: $V = IR$
- e.m.f-1 = $I_1R_1 + I_3R_3$
- 6.0 = $(I_1 \times 10) + (I_3 \times 30)$
- 6.0 = $10I_1 + 30I_3$ (Equation 2) *[1 mark]*

Apply Kirchoff's second law for the potential difference in the bottom loop of the circuit:

- e.m.f-2 = $I_2R_2 + I_3R_3$
- 2.0 = $(I_2 \times 0) + (I_3 \times 30)$
- 2.0 = $30I_3$
- I_3 = 0.07 A *[1 mark]*

Substitute for I_3 into Equation 2:

- 6.0 = $10I_1 + (30 \times 0.07)$
- 6.0 = $10I_1 + 2.1$
- 4 = $10I_1$
- I_1 = 0.39 A *[1 mark]*

Substitute for I_1 and I_3 into equation 1:

- 0.07 = $0.39 + I_2$
- I_2 = −0.32 A *[1 mark]*

[Total: 4 marks]

Remember that you need to have as many equations as unknown currents. Then solve these equations simultaneously. All the energy given out by the battery is used up by the components of the circuit.

Part (c) [2 marks]

A combination of identical resistors each with resistance R has a total resistance of 250 Ω as shown in Fig. 1.2.

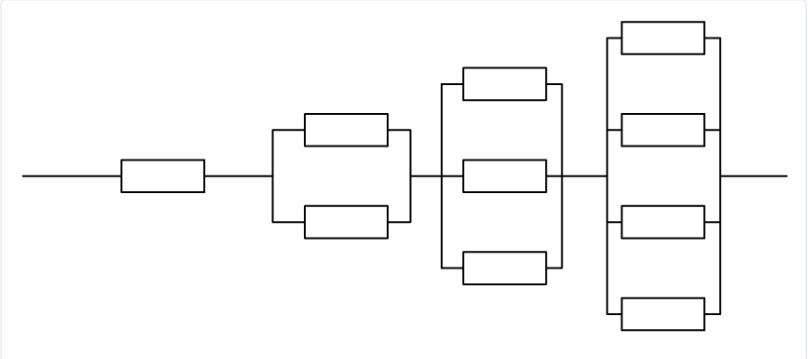


Fig. 1.2

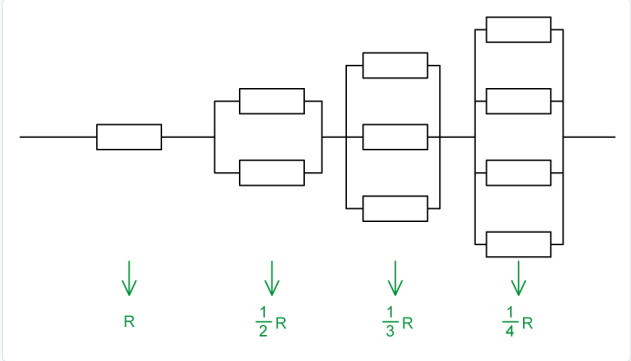
Show, without the use of a calculator, how to find the value of resistance of each resistor and hence determine the value of *R*.

Mark Scheme (2 marks):

Recall the pattern of resistance when identical resistors are placed in parallel:

- Single resistor, $R_{tot} = R$
- Two resistors, $R_{tot} = ½ R$
- Three resistors, $R_{tot} = ⅓ R$
- Four resistors, $R_{tot} = ¼ R$

Under the diagram, write the values of the combinations as fractions:



Write the full equation for resistance in series:

- $total\ resistance = 250 = R + \frac{1}{2}R + \frac{1}{3}R + \frac{1}{4}R$ *[1 mark]*

Write all fractions in terms of a common denominator:

- $250 = \frac{12}{12} R + \frac{6}{12}R + \frac{4}{12}R + \frac{3}{12}R$

Gather the terms:

- $250 = \frac{25}{12}R$

Rearrange and cancel:

- $\frac{250 \times 12}{25} = R$
- $R = 10 \times 12 = 120\ \Omega$ *[1 mark]*

[Total: 2 marks]

*If you went straight to the correct answer by identifying the combinations in terms of fractions of R, and then calculating for a **set of resistors in series**, as shown above, well done!*

But you may have laboriously worked out the value of resistance of each combination in parallel, using $\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$*, this must have taken a long time, and created the opportunity for mistakes to be made.*

It is important to recognise that the sum of three resistors in parallel is given by $3R$ *because:*

$$\frac{1}{R_T} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R}$$

$$\frac{1}{R_T} = \frac{1}{3R}$$

$$R_T = 3R$$

The fact that this answer is only awarded two marks is a clue that there is a quicker way.

*Always pause and think before starting. A moment's thought should be enough to see the pattern of the combinations, and recall the fact that since the resistors are **identical**, a pattern of resistance follows, as described in the solution above.*

Always be ready to apply every part of your knowledge in any situation. This-open minded approach will make you a much better scientist.

Part (d) **[4 marks]**

A student wishes to determine the resistance for three different resistors. They connect the resistors in a 'T' arrangement and change the position of the power source and the voltmeter, as shown in Fig. 1.1**,** to obtain two sets of results, as shown in Table 1.1. The power supply in Circuit 1 causes a current of 0.50 A in one branch of a circuit.

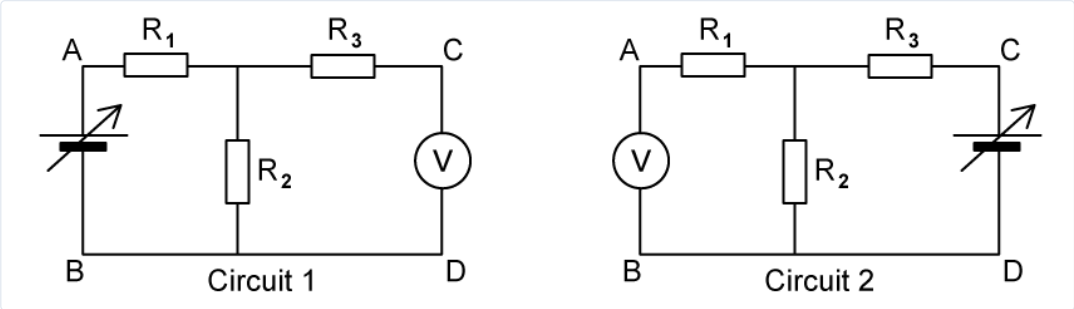


Fig. 1.1

Table 1.1

Circuit	Voltage supplied by power source /V	Reading on voltmeter /V
1	8.0	5.0
2	10	4.0

Calculate the resistance of resistors R_1 , R_2 and R_3 .

You may assume the voltmeter has an infinitely high resistance.

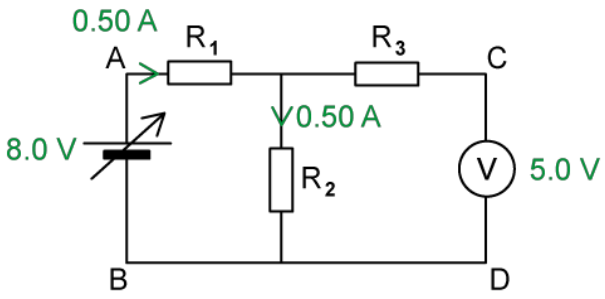
Mark Scheme (4 marks):

List the known quantities:

- Voltage supplied by power source in Circuit 1, V_{S1} = 8.0 V
- Voltage on voltmeter in Circuit 1, V_1 = 5.0 V
- Current flowing through power supply in Circuit 1, I_1 = 0.50 A
- Voltage supplied by power source in Circuit 2, V_{S2} = 10 V
- Voltage on voltmeter in Circuit 2, V_2 = 4.0 V

Consider Circuit 1:

- If the voltmeter has an infinitely high resistance, then current will flow through both R_1 and R_2 :



Calculate the resistance of R_2 :

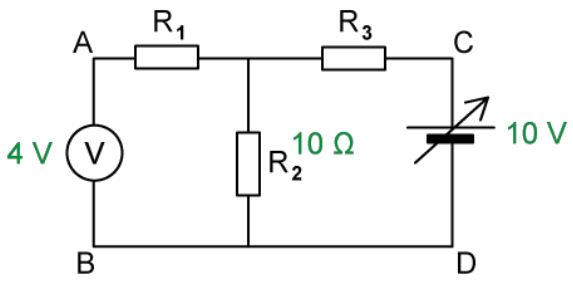
- Recall: *V = IR*
- $R_2 = \frac{V_1}{I_1} = \frac{5.0}{0.5} = 10\ \Omega$ *[1 mark]*

Calculate the resistance of R_1 :

- Voltage across R_1 , $V_{R1} = V_{S1} - V_1 = 8.0 - 5.0 = 3.0$ V
- $R_1 = \frac{V_{R1}}{I_1} = \frac{3.0}{0.5} = 6\ \Omega$ *[1 mark]*

Consider Circuit 2:

- If the voltmeter has an infinitely high resistance, then current will flow through both R_2 and R_3 :



Calculate the current, *I*, flowing through R_2 :

- $I = \frac{V_2}{R_2} = \frac{4}{10}$ = 0.4 A *[1 mark]*

Calculate the resistance of R_3 :

EITHER

- Voltage across R_3 , $V_{R3} = V_{S2} - V_2$ = 10 – 4.0 = 6.0 V
- $R_3 = \frac{V_{R3}}{I} = \frac{6.0}{0.4}$ = 15 Ω *[1 mark]*

OR

- Total resistance of $R_2 + R_3$, $R_{2+3} = \frac{V_{S2}}{I} = \frac{10}{0.4}$ = 25 Ω
- Resistance, $R_3 = R_{2+3} - R_2$ = 25 – 10 = 15 Ω *[1 mark]*

Conclusion:

- R_1 = 6 Ω
- R_2 = 10 Ω
- R_3 = 15 Ω

[Total: 4 marks]

Question 7 Hard [15 marks]

Part (a) [4 marks]

There is current flowing from P to R in the resistor network shown. The potential differences across three of the resistors are 3.0 V, 6.0 V and 5.0 V as shown in Fig. 1.1.

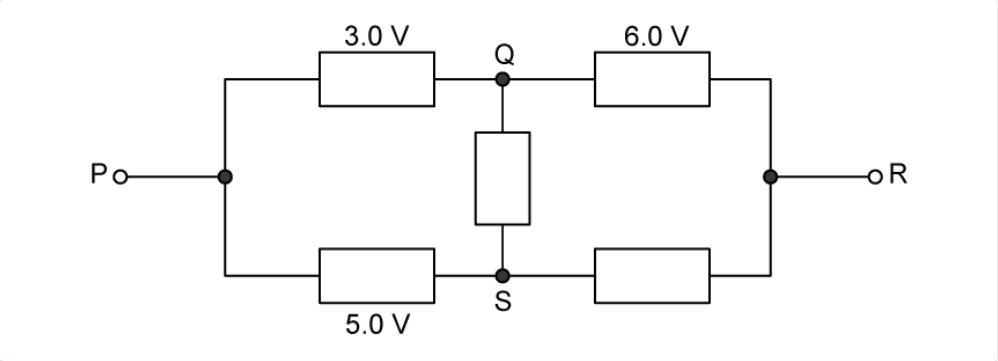


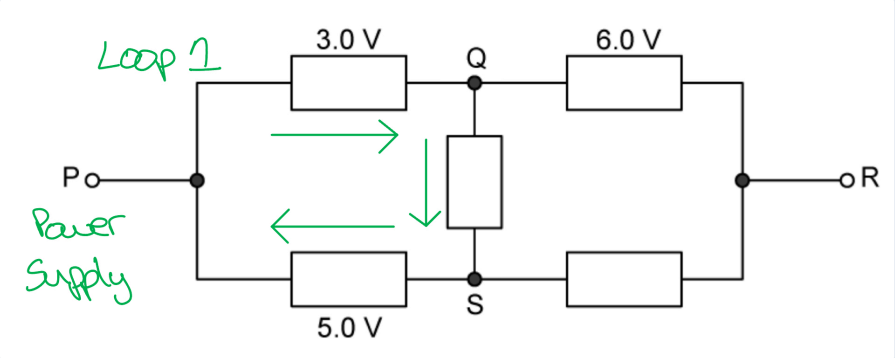
Fig. 1.1

Determine the potential difference across the other two resistors.

Mark Scheme (4 marks):

Draw a diagram to determine the direction of the potential difference in loop 1:

A diagram that would score 1 mark should look like this:



- 3 V potential difference pointing to the right

  AND

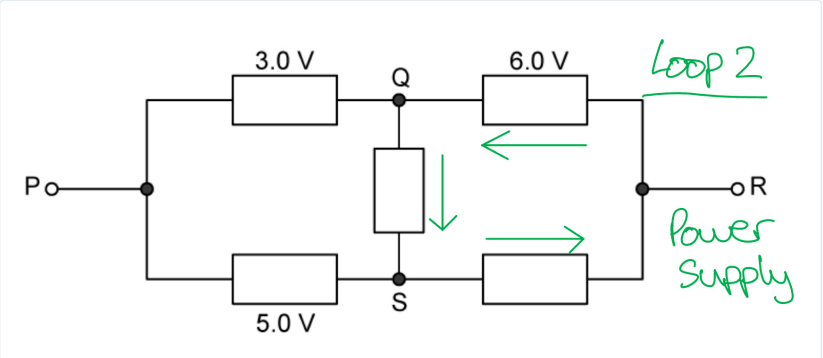
- 5 V potential difference pointing to the left; *[1 mark]*

Calculate the missing potential difference of the resistor, *x*:

- $3 + x + -5 = 0$
- $x = 2$ V *[1 mark]*

Draw a diagram to determine the direction of the potential difference in loop 2:

A diagram that would score 1 mark should look like this:



- A 6 V potential difference pointing to the left

  AND

- A 2 V potential difference pointing from Q to S; *[1 mark]*

Calculate the missing potential difference of the resistor, *y*:

- $-6 + 2 + y = 0$
- $y = 4$ V *[1 mark]*

[Total: 4 marks]

Remember, if there is just one power supply in a circuit, then the resistors can be reduced to a single equivalent resistor, but if there is more than one power supply, then part of one branch can be combined to one resistor. Kirchhoff's laws are then applied to each loop.

Kirchhoff's second law states that the sum of all potential differences around a complete loop is equal to 0.

Part (b) **[4 marks]**

The current through the branch QS is 3.0 A. The total combined resistance of the resistors in the outer loop of the circuit is *r*. The resistance in the outer loop of the circuit is split according to the ratio of the magnitudes of the potential differences of the resistors on each side of the loop.

Obtain an expression, in terms of *r*, for the total resistance of all the resistors in the circuit.

Mark Scheme (4 marks):

List the known quantities:

- Current through QS, I_1 = 3.0 A
- p.d. across QS, V_{QS} = 2.0 V (from part a)
- p.d. across the LHS branch, V_{LHS} = 3 V + 5 V = 8 V
- p.d. across the RHS branch, V_{RHS} = 6 V + 4 V = 10 V (from part a)

Calculate the resistance of the branch QS:

- Recall: *V = IR*
- $R_{QS} = \frac{V_{QS}}{I_1}$
- $R_{QS} = \frac{2}{3}\ \Omega$ *[1 mark]*

Determine the ratio of resistance on each side of the outer loop:

- $V_{LHS} : V_{RHS} = R_{LHS} : R_{RHS}$
- 8 : 10 = 4 : 5 *[1 mark]*

Calculate the LHS and RHS resistances as fractions of *r*:

- $R_{LHS} = \frac{4}{9}r$ **AND** $R_{RHS} = \frac{5}{9}r$ *[1 mark]*

Calculate the total resistance of all the resistors in the circuit:

- The QS branch is the single shared resistor between the two loops, so it is counted once
- $R = R_{LHS} + R_{RHS} + R_{QS}$
- $R = \frac{4}{9}r + \frac{5}{9}r + \frac{2}{3}$
- $R = r + \frac{2}{3}\ \Omega$ *[1 mark]*

[Total: 4 marks]

The two power supplies mean the resistors on each side of the outer loop carry different currents, but the ratio given in the question lets you partition r without solving for the individual resistor values.

The QS branch is a single resistor in the network. When summing the resistances of the circuit, count it once — it does not appear in both loops separately.

This question is probably harder than anything you will encounter in your exam, but that makes it great practice!

Part (c) **[7 marks]**

Diagram 1.2 shows a circuit where V_1 is a battery of 0.5 V, V_2 a battery of 1 V and V_3 a battery of 2 V. Each resistor has a resistance of 0.25 Ω.

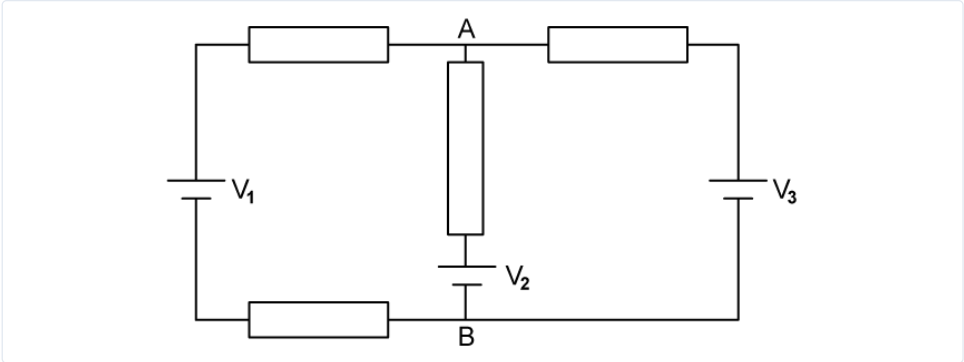


Fig. 1.1

Determine the value of each current at junction A.

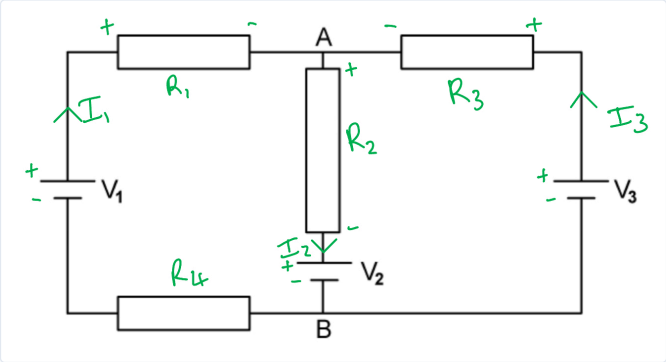
Mark Scheme (7 marks):

List the known quantities:

- V_1 = 0.5 V
- V_2 = 1 V
- V_3 = 2 V
- Resistance of each resistor, R = 0.25 Ω

Label the diagram to show the loops, polarity of the batteries and directions of each current:

A diagram that would score 1 mark should look like this:



- Currents I_1, I_2 and I_3 are labelled as shown with arrows in the correct direction

AND

- Positive and negative battery and resistor terminals should be indicated correctly with the positive side of the resistor the side where the current enters; *[1 mark]*

Use Kirchhoff's first law to obtain an equation for the currents at junction A:

- $I_1 + I_3 = I_2$ *[1 mark]*

Use Kirchhoff's second law to obtain an equation for the sum of the potential differences in loops 1 and 2:

- Recall Ohm's Law: $V = IR$
- Loop 1:
  - $V_1 - I_1R_1 - I_2R_2 - V_2 - I_1R_4 = 0$
  - $0.5 - (0.25I_1) - (0.25I_2) - 1 - (0.25I_1) = 0$
  - $-0.5 - 0.5I_1 - 0.25I_2 = 0$
  - $0.5I_1 + 0.25I_2 = -0.5$

AND

- Loop 2:
  - $V_3 - I_3R_3 - I_2R_2 - V_2 = 0$

- $2 - (0.25I_3) - (0.25I_2) - 1 = 0$
- $1 - (0.25I_3) - (0.25I_2) = 0$
- $1 = 0.25I_3 + 0.25I_2$
- $I_3 + I_2 = 4$ *[1 mark]*

Substitute for I_3 into the equation for current:

- From loop 2: $I_3 = 4 - I_2$
- Equation for current: $I_1 + I_3 = I_2$
- So, $I_1 + (4 - I_2) = I_2$
- $I_1 = 2I_2 - 4$ *[1 mark]*

Now substitute for I_1 from above into the equation for loop 1:

- From loop 1: $0.5I_1 + 0.25I_2 = -0.5$
- $0.5(2I_2 - 4) + 0.25I_2 = -0.5$
- $I_2 - 2 + 0.25I_2 = -0.5$
- $1.25I_2 = -0.5 + 2$
- $I_2 = 1.2$ A *[1 mark]*

Now substitute for I_2 to obtain I_1:

- $I_1 = (2 \times 1.2) - 4$
- $I_1 = -1.6$ A *[1 mark]*

Substitute for I_2 into the equation for loop 2:

- From loop 2: $I_3 + I_2 = 4$
- $I_3 = 4 - 1.2$
- $I_3 = 2.8$ A *[1 mark]*

[Total: 7 marks]

You can label your circuit in any way but make sure the direction of the current is consistent all the way around.

When obtaining an equation for the sum of the potential differences if you cross from:

- *− terminal of one resistor to the + terminal of another resistor* $\rightarrow$ *the change in potential difference is +*
- *+ terminal of one resistor to the − terminal of another resistor* $\rightarrow$ *the change in potential difference is −*

It is always a good idea to check your answer at the end. Here you can do this by using the equation for current:

- $I_1 + I_3 = I_2$
- *So, −1.6 + 2.8 = 1.2 which is correct*